

A sharp Hilbert-core obstruction at $p > 2$ for Shkarin's coanalytic Toeplitz question

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Abstract

Question 5.4 of Shkarin asks whether an expanding coanalytic Toeplitz operator can be weakly hypercyclic on H^p for some $p > 2$. We do not settle the existence question. We prove, however, that the Hilbertian mechanism behind the positive H^2 theorem cannot work above the endpoint under *any* choice of invariant Hilbert core. Precisely, if T is the coanalytic Toeplitz operator induced by a nonconstant $g \in H^\infty$ with $g(\mathbb{D}) \cap \mathbb{D} = \emptyset$, U is a Hilbert-space isometry without nonzero finite-dimensional invariant subspaces, and a bounded map J satisfies $TJ = JU$, then $J = 0$. The proof first identifies every bounded forward orbit with a boundary functional in L^p on the contact set. Wold decomposition then reduces the shift part to an impossible $H^2 \rightarrow L^p$ boundary embedding and the unitary part to an impossible non-atomic $L^2 \rightarrow L^p$ embedding. The exponent contradiction is exactly $p/2 > 1$.

1 Source question and status

Let M_g denote multiplication by g on a Hardy space. In the Hilbert case its adjoint is the coanalytic Toeplitz operator T_g^* . Shkarin proves weak hypercyclicity on H^2 when $g(\mathbb{D}) \cap \mathbb{D} = \emptyset$ and the boundary contact set $\{|g| = 1\}$ has positive measure. After observing that the proof extends by comparison of topologies to $1 \leq p \leq 2$, the paper asks:

Question 5.4. Is there any $g \in H^\infty(\mathbb{D})$ with $g(\mathbb{D}) \cap \mathbb{D} = \emptyset$ such that T_g^* is weakly hypercyclic as an operator on $H^p(\mathbb{D})$ for some $p > 2$?

$$T = S \oplus (-S) \in L(\mathcal{H} \times \mathcal{H}).$$

That is, $T(u, v) = (Su, -Sv)$ for every $u, v \in \mathcal{H}$. By Corollary 5.2 T is 1-weakly hypercyclic. On the other hand, $T^2 = S^2 \oplus S^2 = (S \oplus S)^2$. Thus the non-cyclicity of $S \oplus S$ implies that T^2 is non-cyclic. $\square$

Theorems 1.9 and 1.10 leave the following gap.

Question 5.3. Let $g \in H^\infty(\mathbb{D})$ be such that $|g| > 1$ almost everywhere on $\mathbb{T}$ and $\log(|g| - 1) \notin L^1(\mathbb{T})$. Is it possible for $T_g^* \in L(H^2(\mathbb{D}))$ to be weakly hypercyclic or at least 1-weakly hypercyclic?

Note that the standard comparing the topologies argument extends Theorem 1.9 to coanalytic Toeplitz operators acting on $H^p(\mathbb{D})$ with $1 \leq p \leq 2$. Furthermore, the proof of Theorem 1.10 can be adjusted to work for coanalytic Toeplitz operators acting on every $H^p(\mathbb{D})$ for $1 \leq p < \infty$. In this article we just preferred to stick to the Hilbert space situation. It seems though that our proof of Theorem 1.9 fails miserably for coanalytic Toeplitz operators acting on $H^p(\mathbb{D})$ with $p > 2$.

Question 5.4. Is there any $g \in H^\infty(\mathbb{D})$ with $g(\mathbb{D}) \cap \mathbb{D} = \emptyset$ such that T_g^* is weakly hypercyclic as an operator on $H^p(\mathbb{D})$ for some $p > 2$?

5.1 $2I + S$ is not 1-weakly hypercyclic if $\|S\| \leq 1$

Fix $2 < p < \infty$ and let $q = p/(p-1) \in (1, 2)$. Under standard Hardy duality, the coanalytic Toeplitz operator on H^p is conjugate-linearly isometrically similar to the Banach adjoint

$$T = M'_g \in \mathcal{L}((H^q)^*).$$

This avoids using Hilbert-adjoint notation outside $p = 2$. Conjugate-linear similarity preserves weak density; when an intertwining Hilbert space occurs, one passes to its conjugate complex structure.

Theorem 1 (No isometric Hilbert core above the endpoint). *Let $2 < p < \infty$, $q = p/(p-1)$, and let $g \in H^\infty(\mathbb{D})$ be nonconstant with $g(\mathbb{D}) \cap \mathbb{D} = \emptyset$. Put $X = (H^q)^*$ and $T = M'_g$ on X . Let U be an isometry on a Hilbert space $\mathcal{H}$ having no nonzero finite-dimensional invariant subspace. If $J : \mathcal{H} \rightarrow X$ is bounded and*

$$TJ = JU,$$

then $J = 0$.

Corollary 2 (Obstruction to the source criterion). *There is no nonzero T -invariant subspace $F \subseteq X$ carrying an inner product for which*

- (i) *the Hilbert norm is stronger than the norm inherited from X , and*
- (ii) *$T|_F$ extends to an isometry without nonzero finite-dimensional invariant subspaces on the Hilbert completion of F .*

In particular, conditions (B1)–(B2) of Theorem 1.2 in the source paper can never hold for this operator on H^p when $p > 2$.

2 Idea of the proof

Write $h = 1/g$. It is an analytic self-map of the disc. If $x \in X$ has a bounded forward T -orbit, then

$$x(f) = (T^n x)(h^n f).$$

As $n \rightarrow \infty$, the H^q norm of $h^n f$ loses all boundary mass except on $A = \{|h| = 1\}$. Thus x is represented by an $L^p(A)$ boundary density. Every vector in the range of an intertwiner J has a bounded orbit, because U is an isometry. Hence J factors through a bounded map $K : \mathcal{H} \rightarrow L^p(A)$ intertwining U with multiplication by the unimodular function $g|_A$.

The Wold decomposition leaves two cases. A wandering vector in the shift part has an orthogonal orbit, so K would turn the H^2 norm of analytic polynomials into an L^p norm against a nonzero boundary measure. Hardy peak functions force that measure to satisfy $\nu(I) \lesssim |I|^{p/2}$ on every arc; a partition into equal arcs gives $\nu = 0$. On the unitary part, spectral calculus would yield $\|b\|_{L^p(\nu)} \lesssim \|b\|_{L^2(\mu)}$ with μ non-atomic. Partitioning μ into equal-mass pieces again gives $\nu(\mathbb{T}) \lesssim N^{1-p/2} \rightarrow 0$. Both Wold components vanish.

3 Two concentration lemmas

Normalized analytic peak functions give the endpoint obstruction in its cleanest form.

Lemma 3 (Boundary exponent collapse). *Let $0 < r < s < \infty$ and let ν be a finite positive Borel measure on $\mathbb{T}$. If*

$$\|P\|_{L^s(\nu)} \leq C\|P\|_{H^r} \quad \text{for every analytic polynomial } P,$$

then $\nu = 0$.

Proof. Let $I \subset \mathbb{T}$ be an arc of normalized length $\delta \leq 1/2$, centered at ζ , and take $a = (1 - \delta)\zeta$. The analytic function

$$F_{a,r}(z) = \frac{(1 - |a|^2)^{1/r}}{(1 - \bar{a}z)^{2/r}}$$

uses the analytic branch of the power in the disc. Its boundary modulus to the r th power is the Poisson kernel, hence $\|F_{a,r}\|_{H^r} = 1$. Moreover $|F_{a,r}| \geq c_r \delta^{-1/r}$ on I . Since $F_{a,r}$ is analytic beyond the closed disc, its Taylor polynomials converge uniformly there; the assumed inequality therefore applies to $F_{a,r}$. Consequently

$$\nu(I)^{1/s} c_r \delta^{-1/r} \leq C, \quad \text{so} \quad \nu(I) \leq C_{r,s} \delta^{s/r}.$$

Partition $\mathbb{T}$ into N equal arcs. Since $s/r > 1$,

$$\nu(\mathbb{T}) \leq C_{r,s} N N^{-s/r} \longrightarrow 0.$$

Thus $\nu = 0$. □

Lemma 4 (Non-atomic $L^2 \rightarrow L^p$ collapse). *Let $p > 2$, let μ be a finite non-atomic positive measure on $\mathbb{T}$, and let ν be another finite positive measure. If*

$$\|b\|_{L^p(\nu)} \leq C \|b\|_{L^2(\mu)} \tag{3.1}$$

for every Laurent polynomial b , then $\nu = 0$.

Proof. Uniform density of Laurent polynomials in $C(\mathbb{T})$ first extends (3.1) to continuous functions. This implies $\nu \ll \mu$: if a compact E has $\mu(E) = 0$, choose continuous functions equal to one on E and supported in open sets of arbitrarily small μ -measure. Regularity then handles all μ -null sets. Approximation in $L^2(\mu) + L^p(\nu)$ extends (3.1) to indicators of Borel sets:

$$\nu(E)^{1/p} \leq C \mu(E)^{1/2}. \tag{3.2}$$

For each N , non-atomicity supplies a measurable partition $\mathbb{T} = E_1 \sqcup \cdots \sqcup E_N$ with $\mu(E_j) = \mu(\mathbb{T})/N$. Summing (3.2) to the p th power gives

$$\nu(\mathbb{T}) \leq C^p \mu(\mathbb{T})^{p/2} N^{1-p/2} \longrightarrow 0.$$

□

4 Bounded orbit vectors live on the contact set

Put $h = 1/g$. The hypothesis and the open mapping theorem give $|g(z)| > 1$ in $\mathbb{D}$, so $h \in H^\infty$, $\|h\|_\infty \leq 1$, and $|h(z)| < 1$ in $\mathbb{D}$. Let

$$A = \{\zeta \in \mathbb{T} : |h(\zeta)| = 1\}, \quad R : H^q \longrightarrow L^q(A), \quad Rf = f|_A.$$

Lemma 5 (Contact factorization of bounded orbits). *If $x \in X$ and $\sup_{n \geq 0} \|T^n x\| \leq M$, then there is a unique $\varphi \in L^p(A)$ such that $x = R'\varphi$, and $\|\varphi\|_{L^p(A)} \leq M$.*

Proof. For $f \in H^q$ and $n \geq 0$,

$$x(f) = (T^n x)(h^n f),$$

because $g^n h^n = 1$. Therefore

$$|x(f)| \leq M \|h^n f\|_{H^q}.$$

Dominated convergence on the boundary gives

$$\lim_{n \rightarrow \infty} \|h^n f\|_{H^q} = \left(\int_A |f|^q dm \right)^{1/q} = \|Rf\|_{L^q(A)}. \quad (4.1)$$

If $m(A) = 0$, this already gives $x = 0$. Otherwise, R is injective by Hardy boundary uniqueness. Its range is dense in $L^q(A)$. Indeed, an $L^p(A)$ annihilator, extended by zero to $\mathbb{T}$, has one-sided vanishing Fourier coefficients and is therefore the boundary function of a Hardy function (up to conjugation); it vanishes on $\mathbb{T} \setminus A$ and hence is zero by the same uniqueness theorem. Here $m(\mathbb{T} \setminus A) > 0$: otherwise h would be an invertible inner function, hence a unimodular constant, contrary to the nonconstancy of g .

It follows from (4.1) that x defines a bounded functional of norm at most M on the dense range of R . Extend it to $L^q(A)$ and use the linear bilinear-pairing realization $L^q(A)' = L^p(A)$. This gives $x = R'\varphi$ and the stated norm bound. Density gives uniqueness. $\square$

Using the bilinear pairing, direct calculation gives

$$TR'\varphi = R'(u\varphi), \quad u = g|_A, \quad |u| = 1 \text{ a.e. on } A. \quad (4.2)$$

Here $R'\varphi(f) = \int_A f\varphi dm$.

5 Proof of the main theorem

For $\xi \in \mathcal{H}$,

$$\sup_{n \geq 0} \|T^n J\xi\| = \sup_{n \geq 0} \|JU^n \xi\| \leq \|J\| \|\xi\|.$$

Lemma 5, including uniqueness, therefore produces a bounded linear map $K : \mathcal{H} \rightarrow L^p(A)$ such that

$$J = R'K, \quad \|K\| \leq \|J\|.$$

Using $TJ = JU$, (4.2), and injectivity of R' gives

$$KU = M_u K. \quad (5.1)$$

Apply the Wold decomposition $\mathcal{H} = \mathcal{H}_u \oplus \mathcal{H}_s$, where U is unitary on $\mathcal{H}_u$ and a unilateral shift (of arbitrary multiplicity) on $\mathcal{H}_s$.

First let ξ belong to the wandering subspace of the shift part and put $\varphi = K\xi$. The vectors $U^n \xi$ are orthogonal. Hence, for every analytic polynomial P ,

$$\begin{aligned} \|P(u)\varphi\|_{L^p(A)} &= \|KP(U)\xi\|_{L^p(A)} \\ &\leq \|K\| \|P(U)\xi\|_{\mathcal{H}} = \|K\| \|\xi\| \|P\|_{H^2}. \end{aligned}$$

Push the measure $|\varphi|^p dm|_A$ forward by $u : A \rightarrow \mathbb{T}$ and call the result ν . The last display is exactly $\|P\|_{L^p(\nu)} \leq C\|P\|_{H^2}$. Lemma 3, with $r = 2$ and $s = p$, gives $\nu = 0$, so $K\xi = 0$. Equation (5.1) then makes K vanish on all of $\mathcal{H}_s$.

Now take $\xi \in \mathcal{H}_u$ and put $\varphi = K\xi$. Let μ_ξ be the spectral measure of ξ for the unitary restriction of U , and again let ν be the pushforward of $|\varphi|^p dm|_A$ by u . Because M_u and $U|_{\mathcal{H}_u}$ are invertible, (5.1) also intertwines their negative powers. Thus, for every Laurent polynomial b ,

$$\|b\|_{L^p(\nu)} = \|b(u)\varphi\|_{L^p(A)} \leq \|K\| \|b(U)\xi\|_{\mathcal{H}} = \|K\| \|b\|_{L^2(\mu_\xi)}.$$

The measure μ_ξ is non-atomic: an atom would produce an eigenvector of U , hence a one-dimensional invariant subspace. Lemma 4 gives $\nu = 0$, so $K\xi = 0$. Therefore $K = 0$ on both Wold summands, and $J = R'K = 0$. This proves Theorem 1.

For Corollary 2, the inclusion of F into X extends, by the stronger-Hilbert-topology assumption, to a bounded nonzero map J from the Hilbert completion of F into X . The extended isometry U satisfies $TJ = JU$, contradicting Theorem 1.

6 What this settles, and what it does not

The theorem strictly strengthens the immediate observation that $L^2(A) \not\hookrightarrow L^p(A)$ for $p > 2$. It rules out weighted or singular contact measures, smooth or Sobolev regularizations, arbitrary unitary spectral models, and also the pure unilateral-shift part of an arbitrary isometric Hilbert model. It therefore proves that the phrase “fails miserably” preceding Question 5.4 reflects a structural endpoint, not a missing estimate in the displayed $L^2(A)$ construction.

It does *not* prove that T is not weakly hypercyclic. Weak closure in an infinite-dimensional Banach space is not sequential: a weakly dense orbit may approach zero only through vectors of unbounded norm. The contact factorization controls bounded forward orbits and all isometric Hilbert cores, but not such genuinely nonsequential behavior. Eight focused upgrade attempts tested smooth-contact series, finite n -weak reductions, Hardy–Sobolev cores, direct interpolation from H^2 , and reciprocal-growth obstructions. Each either falls under Theorem 1 or stops at this nonsequential weak-closure gap. Question 5.4 remains open in this packet.

7 Novelty and verification notes

The run’s registry, solution, attempt, and proof-gap indexes were searched for the arXiv identifier, title, “coanalytic Toeplitz”, “weakly hypercyclic”, and the $p > 2$ formulation. Bounded web searches through 17 August 2026 used the exact Question 5.4 wording and close variants. They found the source, later work on norm/subspace hypercyclic Toeplitz operators, and a 2025 paper on smooth-symbol norm hypercyclicity, but no stated resolution of Question 5.4 and no version of Theorem 1. Novelty confidence is therefore moderate, not definitive.

The proof was audited at the following pressure points: the duality orientation in (4.2); the dominated-convergence identity (4.1); density of boundary restriction; extension of the measure inequalities from polynomials to continuous/Borel tests; atomlessness of unitary spectral measures; and both Wold components. No computational claim is used.

Human-review recommendation. Check especially the range-density sentence in Lemma 5, the bilinear duality convention leading to $u = g|_A$, and the passage from Laurent polynomials to indicators in Lemma 4. These are the only places where a convention or a standard density fact is compressed.

References

- [1] P. Duren, *Theory of H^p Spaces*, Academic Press, 1970.
- [2] S. Shkarin, *Orbits of coanalytic Toeplitz operators and weak hypercyclicity*, arXiv:1210.3191 (2012), especially Theorem 1.2, Theorem 1.9, and Question 5.4.
- [3] H. Wold, *A Study in the Analysis of Stationary Time Series*, Almqvist & Wiksell, 1938; see the standard Wold decomposition for Hilbert space isometries.